

BELEM, Brazil

WMO: 821930

Lat: 1.379S

Lon: 48.476W

Elev: 54

StdP: 14.67

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	73.0	73.2	67.8	102.2	83.2	68.9	106.5	81.6	22.0	81.4	18.7	82.3	5.1	100	0.352

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	14.2	91.8	78.5	91.3	78.4	90.0	78.3	82.5	86.7	81.8	86.4	81.2	86.2	8.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.0	161.4	85.0	80.7	159.8	84.9	80.2	157.2	84.5	46.4	87.4	45.8	86.6	45.1	86.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	15.6	13.7	DB	71.2	94.1	1.1	1.5	70.4	95.2	69.8	96.1	69.1	97.0	68.3	98.1	(4)
(5)			WB	69.2	83.2	1.9	1.8	67.8	84.5	66.7	85.6	65.6	86.7	64.2	88.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.7	81.0	80.5	80.6	81.2	81.9	82.1	82.1	82.2	82.1	82.1	82.5	82.0
	DBStd	1.69	1.94	2.10	1.87	1.79	1.65	1.20	1.29	1.36	1.24	1.19	1.33	1.56
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11575	962	855	950	937	989	964	996	997	964	996	974	991
	CDD65	6100	497	435	485	487	524	514	531	532	514	531	524	526
	CDH74	55480	4129	3354	3793	4015	4620	4812	5053	5118	4970	5311	5394	4911
	CDH80	18661	1175	858	928	1069	1470	1784	1953	1952	1855	1970	2008	1639
Wind	WSAvg	6.4	5.9	5.8	5.6	5.3	5.5	6.4	6.6	7.2	7.5	7.4	7.4	6.6
Precipitation	PrecAvg	119.4	14.7	15.1	18.3	16.1	12.2	7.6	6.9	6.2	4.7	4.4	4.3	8.9
	PrecMax	144.3	20.3	22.2	24.8	21.5	18.2	11.6	12.3	9.2	7.6	8.1	10.9	17.3
	PrecMin	100.4	9.3	10.1	10.0	11.4	7.1	2.4	2.5	3.0	1.5	0.6	0.8	2.7
	PrecStd	12.8	2.9	3.0	3.2	2.6	2.7	2.0	2.1	1.7	1.5	2.1	2.3	3.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.4	91.6	90.9	91.3	91.8	91.7	91.8	93.1	92.8	92.5	93.0	91.8
		MCWB	79.6	79.9	79.7	80.2	80.0	78.3	77.5	77.9	78.8	78.3	78.7	78.4
	2%	DB	89.3	89.0	88.1	89.3	90.0	90.4	91.1	91.3	91.0	90.6	91.0	89.9
		MCWB	79.1	78.9	79.4	80.1	79.2	78.0	77.3	78.0	78.2	78.2	78.3	78.5
	5%	DB	87.6	87.1	86.3	87.6	89.2	89.6	89.8	89.7	89.5	89.3	89.4	88.2
		MCWB	79.1	79.1	79.1	79.8	79.1	77.7	77.4	78.1	78.2	78.3	78.4	78.8
	10%	DB	85.9	85.3	84.6	85.9	87.5	88.1	88.2	88.1	87.9	87.8	87.9	87.4
		MCWB	78.9	78.6	78.5	79.2	79.2	78.0	77.8	78.5	78.6	78.8	78.9	78.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.3	82.4	82.2	83.1	82.7	81.9	81.8	82.3	82.4	82.5	82.6	82.8
		MCDB	86.2	86.8	85.8	87.1	87.3	86.8	86.8	86.9	86.7	86.5	86.7	86.5
	2%	WB	81.3	81.0	81.1	81.7	81.6	80.8	80.7	81.1	81.1	81.4	81.5	81.5
		MCDB	85.7	85.4	85.0	85.8	86.6	86.6	86.7	86.9	86.7	86.5	86.6	86.3
	5%	WB	80.4	79.9	80.1	80.7	80.6	79.9	79.9	80.1	80.3	80.5	80.6	80.5
		MCDB	85.1	84.2	84.1	84.8	86.1	86.0	86.2	86.5	86.5	86.5	86.6	86.1
	10%	WB	79.4	79.0	79.2	79.7	79.7	79.1	79.0	79.4	79.4	79.6	79.6	79.5
		MCDB	83.6	83.0	82.8	83.6	84.8	85.3	85.7	85.8	85.7	85.7	85.7	84.8
Mean Daily Temperature Range	5% DB	MDBR	12.3	11.7	11.4	11.7	12.8	14.2	14.7	14.8	14.8	14.5	14.2	13.5
		MCDBR	13.8	13.6	12.7	13.2	14.3	15.1	15.5	15.7	15.6	15.4	15.1	14.3
		MCWBR	5.9	5.8	5.7	5.8	5.7	5.6	5.9	6.4	6.4	6.4	6.2	6.0
	5% WB	MCDBR	12.7	12.5	11.9	12.4	13.3	14.3	14.8	15.0	15.1	14.8	14.4	14.0
		MCWBR	6.3	6.1	6.0	6.2	6.0	6.2	6.4	6.8	6.8	6.8	6.4	6.5
Clear-Sky Solar Irradiance	taub	0.416	0.422	0.425	0.403	0.410	0.405	0.416	0.435	0.494	0.489	0.490	0.467	
	taud	2.486	2.470	2.467	2.551	2.508	2.518	2.473	2.395	2.196	2.218	2.218	2.310	
	Ebn at Noon	291	289	286	285	276	273	272	272	263	267	267	274	
	Edh at Noon	36	37	37	33	33	32	34	38	48	47	47	43	
All-Sky Solar Radiation	RadAvg	1475	1379	1423	1492	1578	1702	1782	1880	1894	1830	1765	1583	
	RadStd	93	96	98	79	106	62	49	46	53	75	67	97	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	-1.66	N/A	-0.72	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page